Fixed-Spectrum Swaps Separate Attention Capacity from Assignment

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Abstract

Interpretability often reads attention mass on a gold token as evidence that a transformer routes information to it. We show that this inference is unsafe in a specific, testable way. Statistics of the attention weights alone, including entropy, the maximum, every norm, and the sorted spectrum, are invariant to which token holds which weight, so they cannot identify assignment; and attention mass on the gold token, which does identify assignment, still does not determine the causal effect of that assignment on the output. We introduce the fixed-spectrum swap (FSS), a causal intervention that preserves every permutation-invariant property of an attention row while changing its assignment to task-labeled positions. Across controlled retrieval, pretrained language models, and context-grounded question answering, FSS separates selective capacity from correct assignment and reveals assignment-responsive, saturated, and selector-mismatch regimes. We treat FSS as a controlled instrument and scope it accordingly: it measures sensitivity to a counterfactual reassignment rather than to spontaneous routing, its pretrained effect magnitudes are seed-sensitive and reported as directional (one replicated condition varied $45\times$ across seeds), and outcome-aware circuit selection beats outcome-blind selection without privileging any particular utility score. Within that scope, and most cleanly for single-token retrieval, attention assignment at a fixed spectrum is a causal lever on the prediction, and FSS gives a controlled test of whether a specified post-softmax reassignment of attention changes a model's output.

1 Introduction

Interpretability claims about transformers often rest on how attention is distributed. Attention weights are read as an explanation of which tokens a prediction depends on [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019], and attention mass on a gold token, or a summary of it such as low entropy or a large maximum, is taken as evidence that the model routes information to that token. We ask whether that inference is licensed, and we separate two versions of it. Summary statistics of the weights alone cannot even identify assignment; attention mass on the gold token can, yet still may not tell us whether the routing causally drives the prediction.

The first gap is structural. Entropy, the maximum weight, every norm, and the full sorted spectrum are invariant to permutations of token positions, so two attention rows can share all of them while one places its largest weight on the required evidence and the other on a distractor. A measurement equal on both rows cannot distinguish them. Attention mass on the gold token, m_S, is the sharper and more common measure: it is not permutation-invariant and does pin down the assignment. Our question is therefore not about m_S as a descriptor but about its causal warrant. At a fixed spectrum, does moving mass onto the task-required evidence change the prediction, and is m_S enough to say when it will?

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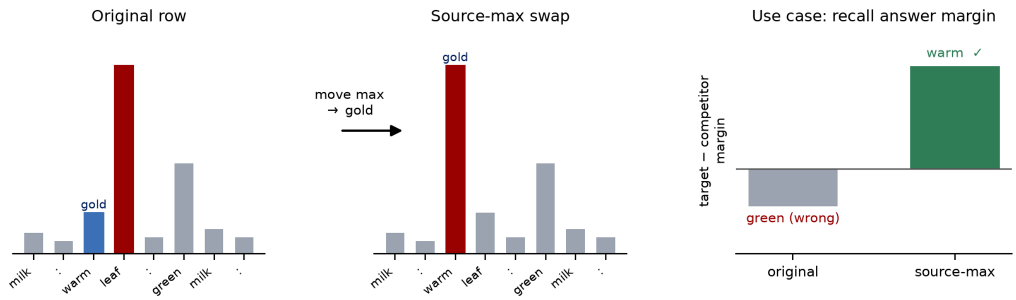

Figure 1: Fixed-spectrum swap (FSS) on in-context key-value recall, one of the conditions we run. *Mechanism* (left, middle): the two rows hold the same sorted weights, and source-max moves the maximum onto the gold value ("warm") while preserving the complete spectrum. *Use case* (right): the paired target-answer margin rises when the maximum lands on the gold value, so the answer switches from a competitor to the gold token; the margin scale is illustrative. The matched control instead swaps two task-irrelevant weights, so treatment and control share spectrum and displacement.

We test this with a fixed-spectrum swap (FSS; Figure 1). At the answer query, we exchange the weight on a known source with a distractor weight and then run the unchanged model. Because the exchange preserves the complete spectrum, a distractor-only exchange gives a matched control. The answer depends on the spectrum's selective capacity, its assignment to task-labeled positions, and the downstream utility of the moved values, and we find that source mass alone does not determine it: selecting circuits by source mass fails where selecting by a value-aware utility term succeeds.

Length is a boundary we test, not a phenomenon we promise to explain. Performance often declines as distractors accumulate [Kazemnejad et al., 2023, Wang et al., 2024, Vasylenko et al., 2025], and attention grows more diffuse while its output variance can collapse [Li et al., 2025]. We use context-length sweeps to ask how far the fixed-spectrum diagnosis remains informative, and we report where it stops rather than claiming it accounts for long-context failure.

Our contributions are:

- **Fixed-spectrum identification.** We separate selective capacity from task-conditioned assignment by preserving the complete attention spectrum and altering only its token placement.
- **Source mass is not sufficient (a methodological observation).** Source mass identifies assignment but does not determine the causal effect: outcome-blind source-mass selection fails where outcome-aware selection succeeds. Because our utility term is the gradient of the measured margin, this is close to definitional and we present it as an observation, not a discovery about attention mass.
- **Transfer and boundaries.** We test the intervention across controlled tasks, pretrained models, a two-required-token task, and natural QA, and identify saturation and selector-mismatch regimes together with the seed, endogenous, selector-circularity, and length boundaries where the reading stops being sufficient.

2 Related Work

Attention weights as routing evidence. Attention weights are widely read as evidence of which tokens a prediction uses, a practice examined in the attention-as-explanation debate [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019, Brunner et al., 2020]. Concentration summaries such as entropy, maximum weight, and output variance are connected to optimization, token selection, and extrapolation [Tarzanagh et al., 2023, Wang et al., 2024, Nakanishi, 2025, Vasylenko et al., 2025, Li et al., 2025, Zsámboki et al., 2025], and move together as context length changes. Our target is not these summaries, which few defend as routing measures, but attention mass on the gold token, which is the measure such claims actually rely on. Mass identifies assignment, so the open question is causal:

does that assignment drive the output, and is mass enough to predict when. We test this while the entire spectrum is held fixed.

Attention intervention and non-identifiability. Attention weights have long been studied as explanations, perturbed to test prediction sensitivity, and shown to be non-identifiable without their value vectors [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019, Brunner et al., 2020, Kobayashi et al., 2020, Hassid et al., 2022]. Causal circuit analysis patches internal activations and measures the resulting output change [Yamakoshi et al., 2023, Conmy et al., 2023]. Attribution patching uses a gradient approximation to screen many candidate components efficiently [Syed et al., 2023, Kramár et al., 2024]. Our utility term has a related first-order form. Here we evaluate it along an exact permutation of a model's own attention row, alongside the finite swap itself.

Closest attention interventions. The nearest methods delete, block, renormalize, or rank parts of an attention pathway: Attention Knockout blocks selected edges and renormalizes the row [Geva et al., 2023], efficient attention projects a row onto its output-effective subspace [Naim and Asher, 2024], and Contribution Weights rank tokens by attention mass combined with projected-value magnitude, testing importance through removal [Cunningham and Muca Cirone, 2026]. Each changes the weight multiset. Our reassignment leaves it intact and changes only its assignment to token positions, and Equation 5 connects that finite, task-labeled swap to the target margin. Sparse and retrieval-based architectures instead change how distant context is accessed [Vasylenko et al., 2025, Hu et al., 2025]; we study a trained row with a fixed spectrum and isolate the effect of assigning its available selectivity to task-required evidence.

3 Methods

3.1 Capacity and Assignment

At an answer query, let $a = (a_1, \ldots, a_n)$ be an attention row over n token positions, where $a_j \geq 0$ and $\sum_j a_j = 1$. Let $S \subseteq \{1, \ldots, n\}$ contain the $r = |S|$ task-required evidence positions, and let $a_{(1)} \geq \cdots \geq a_{(n)}$ denote the same weights in descending order. Define

$$C_r^+ = \sum_{i=1}^{r} a_{(i)}, \qquad C_r^- = \sum_{i=n-r+1}^{n} a_{(i)}, \qquad (1)$$
$$K_r(a) = C_r^+ - C_r^-.$$

C_r^+ and C_r^- are the largest and smallest evidence masses available under permutations of the fixed spectrum, and $K_r(a)$ is the available routing range. An assignment is a permutation π of $\{1, \ldots, n\}$ that places weight $a_{\pi(j)}$ at token position j. Its evidence mass and normalized assignment coordinate are

$$m_S(\pi) = \sum_{j \in S} a_{\pi(j)},$$
$$\rho_S(\pi) = \frac{m_S(\pi) - C_r^-}{K_r(a)} \in [0, 1], \qquad K_r(a) > 0. \qquad (2)$$

Thus $m_S(\pi) = C_r^- + \rho_S(\pi) K_r(a)$. If $K_r(a) = 0$, every assignment has the same evidence mass.

> **Capacity** specifies how much evidence mass the spectrum can support.
> **Assignment** specifies where that mass is placed.
> **Utility** specifies whether moving it changes the answer.

Proposition 1 (Fixed-spectrum capacity-assignment decomposition)**.** *For every assignment π of a fixed weight multiset,*

$$C_r^- \leq m_S(\pi) \leq C_r^+, \qquad (3)$$

and both bounds are attained. If $K_r(a) > 0$, Equation 2 defines a unique $\rho_S(\pi)$. No statistic invariant to token permutation can determine ρ_S.

Proof sketch. An r-position evidence set receives at most the r largest weights and at least the r smallest. Assigning those weights to S attains each endpoint. Normalizing evidence mass within this interval gives ρ_S. Any permutation-invariant statistic is constant across all assignments, including assignments at opposite endpoints. The supplementary material gives the full exchange argument and the corresponding affine margin range. The decomposition is algebraic for every r. Behavioral validation is positive for $r = 1$ and $r = 2$; the $r = 3$ interval includes zero and the $r = 4$ models miss competence, so efficacy at higher arity is unestablished.

3.2 Utility-Conditioned Output Effects

Capacity and assignment describe attention mass. Prediction also depends on values and downstream computation. Let H be the selected attention heads. In each $h \in H$, let s be the evidence position, d_h the donor position whose weight is exchanged with it, and a_{hj} the original attention weight on position j. Define the transferred mass $\delta_h = a_{hd_h} - a_{hs}$. Let x_{hj} be token j's residual-stream contribution after head h's value and output projections. If z is the unpatched residual state, the intervention gives $z' = z + \Delta z$, where

$$\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h}). \tag{4}$$

Let $\ell_k(z)$ be the downstream logit for token k, with target y and competitor c. Define $M_c(z) = \ell_y(z) - \ell_c(z)$, the exact change $\Delta M_c = M_c(z + \Delta z) - M_c(z)$, and local utility $u_{hj}^{(c)} = \nabla_z M_c(z)^\top x_{hj}$.

Proposition 2 (Utility-conditioned swap effect). *If $\nabla_z M_c$ is L_c-Lipschitz on $\{z + t\Delta z : t \in [0, 1]\}$, then*

$$\Delta M_c = \sum_{h \in H} \delta_h \big(u_{hs}^{(c)} - u_{hd_h}^{(c)}\big) + R, \qquad |R| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \tag{5}$$

For an affine downstream margin, $R = 0$.

The Taylor remainder is standard. The new component is the task-labeled capacity-assignment coordinate coupled to the exact FSS vector in Equation 4.

The intervention transfers attention mass from a donor token to the source. It helps when the source contributes more positively to the target margin than the displaced donor, unless downstream curvature overwhelms that gain. Source-max can therefore succeed, saturate, or fail under the same attention-mass rule. The supplementary material derives Equation 5, extends it to evidence sets, and states a sufficient margin certificate.

4 Experimental Setup

Controlled tasks. Argmax retrieval presents key-value pairs and requests the value paired with the largest key. Flag retrieval presents values with exactly one marked position and requests that value. Both have one labeled source. Two-evidence modular sum marks two digits and requests their sum modulo ten, so both sources are required. Four-layer decoders are trained with NoPE and RoPE and four seeds per cell; evaluation reaches $50\times$ training length. An eight-layer, width-512 grid checks scale. The inverse temperature $\beta \in \{1, 2, 4\}$ changes attention capacity while each assignment comparison preserves the resulting sorted spectrum.

Pretrained and natural-language tasks. In-context key-value recall lists pairs followed by a query key; the paired value is the labeled source. We test Qwen2.5, Pythia, Gemma, and SmolLM2 across increasing pair counts. The selector audit compares circuits ranked on disjoint calibration data by source mass or by output-conditioned utility. Natural MCQA uses four-choice questions with one gold passage and unrelated passages. Questions must be correct with the full context and gold passage alone, but incorrect without context. This filtering defines a model-specific context-dependent subset and is not intended to estimate average performance over the full dataset. Its length ladder freezes the four-passage circuit before forming nested 8-, 16-, and 32-passage contexts with additional unrelated passages.

Interventions and estimation. The distractor control swaps the largest and smallest non-source weights, preserving source mass while permuting the same attention row. For a source position s, let $d^+ = \arg\max_j a_j$, $d^- = \arg\min_j a_j$, and $D = \{j : j \neq s\}$ over valid tokens. Source-max swaps (s, d^+) and source-min swaps (s, d^-). In controlled and pretrained key-value recall, the distractor control uses the maximum and minimum within D. Natural QA uses a displacement-matched control. With $\delta = a_{d^+} - a_s$, it chooses

$$(p, q) = \arg \min_{\{j,k\} \subset D} \big| |a_j - a_k| - \delta \big| \tag{6}$$

and swaps (p, q), minimizing the L_1-displacement mismatch to the treatment while holding source mass fixed; every control preserves the complete spectrum. The control does not match the donor token, positional distance, or value norm, which remain separate confounds. Calibration and evaluation examples are disjoint, circuits are frozen before paired evaluation (Figure 2), hierarchical intervals resample calibration or model seeds above paired examples, and a cell is interpreted only after its frozen competence gate passes. Three confirmatory contrasts form one inferential family (controlled source-max versus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max versus matched control), tested with two-sided exact sign-flip tests at the cluster level with Holm correction; other intervals are descriptive and unadjusted. These swaps are counterfactual probes and need not lie on the model's endogenous query-key manifold. Treatment and control share this intervention class, so their contrast measures downstream sensitivity to assignment, rather than the frequency with which the model naturally creates a particular row. Layer controls ($+1.107$ for selected $K = 8$ versus -0.059 for a random layer) and coverage dose response show greater sensitivity under this probe. They do not show that normal query-key computation produces swapped rows or that selected heads are the exclusive retrieval mechanism. Section 5.4 reports the endogenous-row test and the patching comparison that bound this reading.

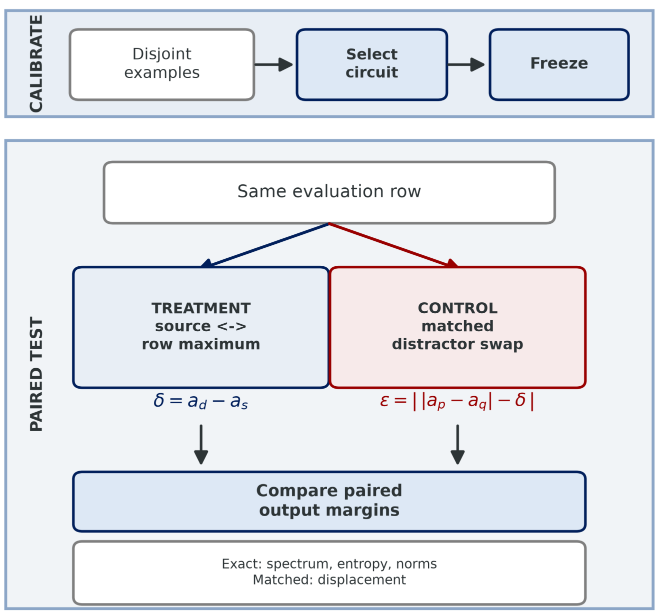

Figure 2: Paired causal protocol. A circuit selected on disjoint calibration data is frozen before treatment and control are applied to the same evaluation row. Both paths preserve the spectrum; the natural QA control additionally minimizes displacement mismatch in Equation 6.

Measured outcomes. We record task accuracy, target-versus-competitor margin, source mass, attention-output variance, entropy, norms, and the full sorted spectrum. Every fixed-spectrum condition preserves the selected attention rows to numerical precision. The supplementary material reports prompts, hyperparameters, per-length tables, competence failures, and preregistered decision rules.

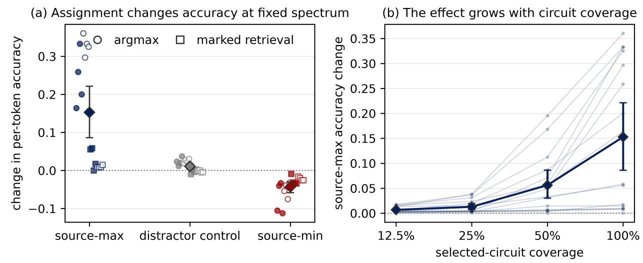

Figure 3: Assignment changes retrieval despite an identical attention spectrum. Source-max improves accuracy, source-min harms it, and the distractor-only control remains near zero (a). The effect increases with the fraction of the selected circuit receiving the intervention (b). Diamonds and intervals summarize trained models; faint paths show individual cells.

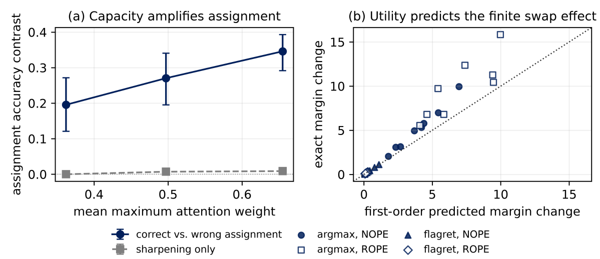

Figure 4: Capacity controls leverage, and utility ranks the finite intervention. Sharper spectra increase the correct-versus-wrong assignment contrast while sharpening alone remains near zero (a). The first-order utility term ranks the exact source-max margin effect (b); panel (b) pools 32 controlled cell means, so it shows the aggregate relation, and the honest per-model correlations (0.87/0.54/0.55) are in Appendix A. Error bars summarize trained models.

5 Results and Discussion

5.1 Assignment Changes Accuracy at Fixed Spectrum

We begin with the simplest test: move the largest available weight to the source, move it away, or exchange two distractors. Across sixteen trained models, the mean changes are $+0.153$, -0.043, and $+0.011$, respectively (Figure 3, with controlled cells in Table 1). Source-max beats the control in every model-length comparison. Their primary cluster-level contrast is $+.142$ (95% CI $[+.080, +.206]$; exact $p = 3.05 \times 10^{-5}$; Holm-adjusted $p = 9.16 \times 10^{-5}$), while complete-spectrum invariant error stays below 10^{-6}.

The effect grows with circuit coverage (Figure 3b): patching one, two, four, and eight selected heads gives mean source-max gains of 0.007, 0.013, 0.056, and 0.153, monotone in 31 of 32 comparisons, consistent with a distributed effect. Source attention tracks accuracy more closely than attention-output variance: normalizing that variance leaves accuracy unchanged, whereas forcing mass onto the known source restores controlled retrieval, and sharpening the original row has little effect. The ordering survives the eight-layer replication, both positional encodings, four seeds, and both controlled tasks, while random-layer and adjacent-layer interventions stay near zero. The full displacement audit, per-cell intervals, and variance and patching results are in the supplementary material.

5.2 Capacity Amplifies Assignment

We next vary spectrum and assignment independently. Temperature scaling changes the maximum available weight, and at every temperature we evaluate identity, source-max, source-min, and distractor control on the same examples. The source-max minus source-min contrast grows from 0.196 to 0.346, while sharpening the identity assignment changes accuracy by at most 0.009 (Figure 4a). Capacity sets the possible leverage, while assignment determines where that leverage lands. The

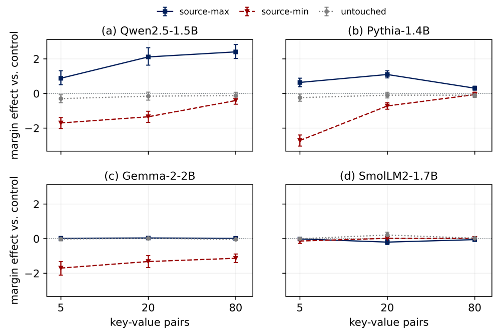

Figure 5: Matched pretrained-model comparison. A single calibration seed (seed 0) is shown because the frozen circuit is selected once per seed and the panels compare conditions within that fixed circuit; seed-pooled results are in the supplementary material. Error bars are paired 95% intervals over $n = 128$ examples per model-length cell. All panels use the 5-, 20-, and 80-pair ladder and report source-max, source-min, and untouched margin relative to the same distractor control. Qwen and Pythia respond positively, Gemma is saturated, and the source-mass-selected SmolLM2 circuit does not benefit.

primary interaction is $+.150$ (95% CI $[+.094, +.210]$; exact $p = 6.10 \times 10^{-5}$; Holm-adjusted $p = 1.22 \times 10^{-4}$).

For held-out examples we differentiate the target margin along the exact source-max path. The first-order term in Proposition 2 ranks the finite margin change, with per-model Spearman correlations of 0.87, 0.54, and 0.55 on Qwen2.5-1.5B, Pythia-1.4B, and Gemma-2-2B and sign agreements of 0.81, 0.81, and 0.71 (Appendix A). Aggregating to the 32 controlled cell means in Figure 4b raises the rank correlation to 0.989, but that pooled figure overstates the per-row relation and we do not use it as the headline. The slope of the actual-on-predicted fit is 1.387, off by about 39%, so we read the utility term as a usable local ranking signal rather than a calibrated estimator of effect size. The remaining error measures downstream curvature along the swap.

Modular sum requires the model to route and combine two marked values. At $5\times$ training length, assigning the top two weights to evidence raises exact match by 0.066, assigning the bottom two lowers it by 0.291, and evidence-max exceeds a control that preserves both the spectrum and total evidence mass by 0.043 (95% CI $[0.017, 0.072]$). This is a small but positive two-evidence effect at $5\times$ (the effect at $1\times$ is $+0.007$), so the result reaches two required tokens rather than one; it does not establish general multi-token routing. A registered three-evidence estimate includes zero, and four-evidence models miss the frozen competence gate, so arity beyond two remains unresolved.

The multi-evidence control preserves both the spectrum and the total evidence mass, changing only which evidence token receives each weight, so its small effect separates aggregate source mass from useful evidence assignment.

5.3 The Same Intervention Produces Three Pretrained Regimes

We read the pretrained results as directional, not calibrated, because the selection procedure itself is seed-unstable. On the one condition we replicated, the Qwen headline seed selects a layer-19 circuit while five further seeds all select a layer-14 circuit, so different seeds recover different circuits, not noisier estimates of one. That headline circuit gives a source-max margin of $+1.882$ on the exact contrast (Table 3) and $+1.990$ over the 5/20/80 ladder in Table 2 (same seed, two aggregations), whereas the five replication seeds sit near $+0.04$: a $45\times$ gap. The six-seed mean $+0.349$ (95% CI $[+.037, +.982]$) excludes zero, so the direction reproduces while the size and the circuit do not, and the single-seed magnitudes in Table 2 are directional only. The robust reading is the sign and the regime pattern: Qwen and Pythia gain margin from source-max, Gemma stays accurate and gains little from extra source mass while source-min still harms it (the saturation regime of Proposition 2), and the source-mass-selected SmolLM2 circuit does not benefit. Gemma's asymmetry ($+0.04$ gain against a -1.37 loss) is not the generic "destruction hurts more than construction helps" pattern, because that pattern is absent in the other models: Qwen and Pythia gain *more* from source-max than

Task	Cond.	Unpatched	Δ src-max	Δ src-min	Δ distr.	**max−distr.**
Single evid. (Acc)	$\beta = 1$	.619	$+.153$	$-.043$	$+.011$	$+.142$
	$\beta = 4$	.619	$+.298$	$-.048$	$+.025$	$+.273$
Two evid. (EM)	$1\times$	.952	$+.019$	$-.559$	$+.012$	$+.007$
	$5\times$	.484	$+.066$	$-.291$	$+.023$	$+.043$

Table 1: Controlled fixed-spectrum assignment. Single-evidence rows report per-token accuracy over $n = 8{,}192$ head-example pairs; two-evidence rows report exact match over $n = 2{,}048$. Source-max moves the row maximum to the source, source-min moves it away, and the distractor control swaps two non-source weights; all preserve the complete spectrum. Full cells and 95% intervals are in the supplementary material.

Model	Unpatched	Δ src-max	Δ src-min	**max−distr.**
Qwen2.5-1.5B†	-3.048	$+1.990$	$-.961$	$+1.799$
Pythia-1.4B	-3.805	$+.826$	-1.022	$+.680$
Gemma-2-2B	$+2.234$	$+.040$	-1.373	$+.027$
SmolLM2-1.7B	-2.782	$-.168$	$-.112$	$-.091$
Output-conditioned circuit selection (ΔM vs. matched control)				
	util. gain	util. gap	src mass	**util−src**
Pythia-1.4B	$+.519$	$+.359$	$+.006$	$+.513$
SmolLM2-1.7B	$+1.253$	$+.859$	$+.006$	$+1.247$
Qwen2.5-1.5B	$+1.493$	$+1.493$	$+.069$	$+1.424$

Table 2: Pretrained models, target-margin units ($n = 384$ per model, matched 5/20/80-pair ladder from Figure 5). **These are single-calibration-seed magnitudes and must not be read as effect sizes; use them only for the sign.** The one condition we replicated (†, Qwen top-block source-max) collapses under reseeding, from the headline $+1.99$ to a six-seed mean of $+0.35$ (median $\approx +0.04$; the per-seed distribution is Table 3), so the other single-seed magnitudes are unquantified and likely inflated. What is robust is the sign pattern (top block) and the regime taxonomy. Bottom: output-conditioned circuit selection; because the utility term is the gradient of this margin, its advantage over source mass (src mass column) shows that outcome-aware beats outcome-blind selection rather than isolating utility (a plain source-gradient selector does slightly better; Section 5.3). A Gemma selector run was not conducted. Full cells and 95% intervals are in the supplementary material.

they lose from source-min. The gain-poor, loss-rich profile is specific to Gemma, the one model already accurate at long context, which is the saturation signature.

SmolLM2's source-mass selector fails, yet ranking heads by an output-conditioned score restores a positive effect. We are careful about what this isolates, because our utility term is the gradient of the very margin we then measure, so selecting heads by predicted ΔM and finding large ΔM is close to guaranteed by Proposition 2. A ten-seed selector ablation separates the two effects. Outcome-aware scores clearly beat outcome-blind ones: source-gradient and utility-gain reach $+0.75$ and $+0.60$ mean margin, while source mass, transfer mass, and a random selector are all near zero or negative (-0.04, -0.09, -0.02). But the utility functional earns no special credit: a plain source-gradient score *beats* it (utility-gain−source-gradient $= -0.15$, 95% CI $[-0.27, -0.03]$), and the preregistered utility-specificity test fails. The supported claim is thus that outcome-aware selection beats outcome-blind selection, not that this particular utility term is the right one. On context-grounded natural QA the same directional pattern holds across six seeds (mean contrast $+.378$, one seed dominant, leave-one-seed-out $+.19$ to $+.55$); we report the interval rather than a significance claim, since an exact sign test at six seeds bottoms out at the $p = .03125$ we reach.

5.4 What the instrument does and does not measure

Two measurements fix the scope of the FSS effect and, read alongside the results above, keep the interpretation precise: FSS is a clean test of a counterfactual reassignment, not a readout of spontaneous routing.

First, the effect is measured on rows the model does not itself produce. On untouched distractor-order variants the conditional source-mass coefficient is $+1.250$ ($p = .03125$), so source mass is associated with the outcome in endogenous data, yet the matched fixed-spectrum contrast on those same rows is $+.026$ ($[-.017, +.070]$, spanning zero). This reconciles the large injected effects in Table 2 (for example $+1.99$ on Qwen) with the null here: source-max is a large off-distribution reassignment while the endogenous rows carry small model-produced differences, so the effect is present in the former and absent in the latter. The consistent reading is that assignment is causally potent when attention is pushed well off what the model would produce, whereas within the model's own distribution source mass on the gold token behaves as a correlate of the query-key computation more than the operative channel.

Second, restoring the selected circuit recovers little of the damage from corrupting it: clean-to-corrupt activation patching on the selected heads rescues $+.308$ margin, only 2.5 to 3.3% of the median corruption damage. Patching and reassignment are not directly comparable (one restores head outputs after corruption, the other acts on an unchanged input), but the small recovered fraction indicates the selected heads are one contribution among several, not the dominant pathway.

5.5 Natural Context Marks the Scope Boundary

At a fixed context in ordinary text, the answer is sensitive to a spectrum-preserving reassignment onto the gold evidence, short of "causal routing to the required evidence" pending the semantic-distractor control (Limitations). The length ladder is a scope test (Figure 6): we select the circuit at four passages, freeze it, and append unrelated passages to form nested 4- to 32-passage contexts. Both Qwen and SmolLM2 lose margin and accuracy by 32 passages, yet their source-mass trajectories diverge: Qwen source mass rises as rescue weakens, while SmolLM2 source mass falls with rescue flat. The shared degradation cannot be pinned to one source-mass trajectory, so this is a scope limit of the diagnosis, not a failed prediction; the fixed-context swap measures whether a chosen circuit can use reassigned evidence, and as contexts grow, representation quality and computation outside the circuit also matter. Exact 32-minus-4 changes are in the supplementary material.

5.6 Identification and Replication Audits

Replication checks vary the calibration sample, patch budget, and model family, and they support the same reading. Selected circuits are positive at every head budget, adjacent and random layers are smaller or near zero, the utility term ranks exact effects within every competent cell while calibrating them imperfectly, and an exact rerun reproduces the direction with an atypically large original magnitude. Appendix A reports these audits in full.

6 Limitations

FSS identifies the causal effect of a specified post-softmax reassignment, not a complete account of model computation: the swap is an off-manifold counterfactual, and preserving the spectrum isolates assignment without showing that training produces the same row. Three comparisons would each strengthen a bounded claim. The *semantic-distractor* control, moving the maximum onto a competitor answer token at matched spectrum and displacement, is the primary one: without it the natural-QA result is compatible with sensitivity to any answer-shaped token rather than to the required evidence. Adding calibration seeds would replace the seed-fragile pretrained point estimates with distributions, and a head-to-head against Attention Knockout and Contribution Weights would make the "leave the multiset intact" distinction empirical. Higher arity is competence-limited and the generation audit is short-horizon. Anonymized code, seeds, prompts, and per-example outputs accompany the submission and will be released on acceptance.

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A Identification and replication audits

Replication checks vary the calibration sample, patch budget, and model family. Across three frozen Qwen calibration seeds, selected circuits are positive at every head budget: $K = 2, 4, 8$ gives $+.460$, $+1.054$, and $+1.107$ margin. Adjacent layers are smaller and inconsistent ($+.147$ and $-.131$), while random layers average near zero ($-.059$). Full intervals and seed standard deviations are reported in the supplementary package.

A separate replication tests the utility relation. Circuits are chosen by source mass on disjoint calibration examples; utility is then computed on held-out evaluation rows. Across three seeds per family, every competent source-max cell is positive, and predicted effects rank the exact effects within each cell. Mean Spearman correlations are $.866$ for Qwen2.5-1.5B, $.544$ for Pythia-1.4B, and $.554$ for Gemma-2-2B, with sign agreements $.805$, $.807$, and $.714$. The approximation is imperfect at large effects, so we use it as a local ranking signal rather than a calibrated finite-effect estimator. For the 32 controlled cell means in Figure 4b, the actual-on-predicted fit is $1.387x - .102$ (Pearson .978), which deviates from identity calibration.

The exact Qwen rerun, with precision, lengths, contrast, and sample counts fixed, is the sharpest evidence that the pretrained magnitudes are seed-sensitive and the main reason we present them as directional. Each seed selects its own circuit before evaluation, and the headline seed selects a different circuit from the other five (Table 3). The sign is reproducible in all six seeds, the six-seed mean source-max minus control margin is $+0.349$ with a hierarchical 95% interval of $[+.037, +.982]$ that excludes zero, but the size varies by more than an order of magnitude. A two-sided exact sign test at six seeds bottoms out at $p = .03125$, so we report the seed-level values rather than a significance verdict. Adding seeds to every pretrained cell, so that Table 2 reports distributions rather than point estimates, is the most important open item, not a cosmetic extension.

Seed	Layer	Heads	Mean margin
0 (headline)	19	$[3, 1, 0, 2]$	$+1.882$
1	14	$[3, 4, 0, 1]$	$+0.041$
2	14	$[3, 4, 0, 1]$	$+0.036$
3	14	$[3, 0, 4, 1]$	$+0.053$
4	14	$[3, 0, 4, 1]$	$+0.039$
5	14	$[3, 0, 4, 1]$	$+0.044$

Table 3: Exact multi-seed Qwen replication. The headline seed selects a layer-19 circuit and a margin $45\times$ the mean of the five replication seeds, which agree on a layer-14 circuit near $+0.04$. The effect is directionally reproducible and magnitude-unstable.

Output-conditioned selectors lead source-mass selection in each family, but this comparison is outcome-aware versus outcome-blind and does not isolate the utility functional, which is the gradient of the measured margin; an outcome-aware but non-utility baseline is not run here. Full selector results, competence-gate failures, and the anonymized code package appear in the supplementary material.

B Theory details

This appendix gives the full statements and proofs behind Propositions 1 and 2, and a multi-evidence corollary. The complete per-cell result tables and secondary figures are in the separate supplementary material.

Established softmax grouping identity. For evidence positions S, distractors D, scores s_j, and temperature $T > 0$, let $Z_G = \sum_{j \in G} e^{s_j/T}$ and $F_G = -T \log Z_G$. Grouping the softmax denominator gives

$$m_S := \sum_{j \in S} a_j = \frac{Z_S}{Z_S + Z_D} = \sigma\left(\frac{F_D - F_S}{T}\right). \tag{7}$$

With the group log-mean-exp score $\bar{s}_G = T \log(|G|^{-1} \sum_{j \in G} e^{s_j/T})$,

$$m_S \geq \alpha \iff \bar{s}_S - \bar{s}_D \geq T\left(\log \tfrac{|D|}{|S|} + \operatorname{logit}(\alpha)\right). \tag{8}$$

These are elementary partition-function identities; we state them only to make the dependence on distractor scores explicit, since distractor count alone is insufficient when arbitrarily many negligible-score distractors need not reduce evidence mass.

Capacity–assignment decomposition. Let $a_{(1)} \geq \cdots \geq a_{(n)}$, let S contain r evidence positions, and define $C_r^+ = \sum_{i=1}^r a_{(i)}$, $C_r^- = \sum_{i=n-r+1}^n a_{(i)}$, and $K_r(a) = C_r^+ - C_r^-$.

Theorem 1 (Capacity–assignment decomposition)**.** *For every permutation π of a fixed attention-weight multiset, $C_r^- \leq m_S(\pi) \leq C_r^+$, both endpoints are attainable, and for $K_r(a) > 0$ the unique coordinate $\rho_S(\pi) = (m_S(\pi) - C_r^-)/K_r(a) \in [0, 1]$ gives $m_S(\pi) = C_r^- + \rho_S(\pi) K_r(a)$. For affine margin $M(\pi) = b + \sum_j a_{\pi(j)} u_j$ with sorted utilities $u_{(1)} \geq \cdots \geq u_{(n)}$,*

$$\max_\pi M(\pi) = b + \sum_i a_{(i)} u_{(i)}, \qquad \min_\pi M(\pi) = b + \sum_i a_{(i)} u_{(n+1-i)}. \tag{9}$$

No statistic invariant to token permutation can determine evidence assignment or the output margin.

Proof. An r-position set receives at most the r largest weights and at least the r smallest; assigning those weights to S attains the endpoints. Normalizing the bounded mass gives ρ_S. For the output range, suppose $u_p > u_q$ but $a_{\pi(p)} < a_{\pi(q)}$. Swapping those weights changes the margin by $(a_{\pi(q)} - a_{\pi(p)})(u_p - u_q) > 0$. Repeated exchanges align weights and utilities for the maximum and reverse them for the minimum, the classical rearrangement inequality [Hardy et al., 1952]. The extremal assignments preserve the complete weight multiset and therefore every permutation-invariant statistic. ☐

Nonlinear downstream network. Fix an input and selected heads H at one layer. Let $x_{hj} = O_h v_{hj}$ and $z = \sum_{h \in H} \sum_j a_{hj} x_{hj}$. For target y and competitor c, let $M_c(z) = g_y(z) - g_c(z)$. A source–donor swap has $\delta_h = a_{hd_h} - a_{hs}$ and $\Delta z = \sum_{h \in H} \delta_h (x_{hs} - x_{hd_h})$. Define local utility $u_{hj}^{(c)} = \nabla M_c(z)^\top x_{hj}$.

Theorem 2 (Nonlinear routing-margin bound)**.** *If ∇M_c is L_c-Lipschitz from z to $z + \Delta z$, then*

$$\left| \Delta M_c - \sum_{h \in H} \delta_h (u_{hs}^{(c)} - u_{hd_h}^{(c)}) \right| \leq \frac{L_c}{2} \|\Delta z\|_2^2. \tag{10}$$

Proof. The swap changes head h by exactly $\delta_h(x_{hs} - x_{hd_h})$. The fundamental theorem of calculus gives $\Delta M_c = \int_0^1 \nabla M_c(z + t\Delta z)^\top \Delta z \, dt$. Subtracting $\nabla M_c(z)^\top \Delta z$ and applying Lipschitzness bounds the remainder by $\int_0^1 L_c t \|\Delta z\|_2^2 \, dt = L_c \|\Delta z\|_2^2 / 2$ [Nesterov, 2004]. Expanding the first-order term gives Equation 10. ☐

Corollary 3 (Multi-evidence routing)**.** *Let S contain r evidence positions and pair every evidence position missing a top-r weight with one top-r non-evidence donor. For disjoint swap pairs P_h and $\delta_{hsd} = a_{hd} - a_{hs} \geq 0$, $\Delta z = \sum_{h \in H} \sum_{(s,d) \in P_h} \delta_{hsd}(x_{hs} - x_{hd})$, and Equation 10 holds with the first-order sum extended over $(s, d) \in P_h$. Assigning top-r capacity to the evidence improves the margin whenever the summed evidence–donor utility gain exceeds the curvature remainder.*

The decomposition and corollary are algebraic for arbitrary r. Their behavioral premise is established for $r = 1$ and $r = 2$; the three-evidence interval includes zero and the four-evidence models miss the competence gate, so efficacy beyond $r = 2$ remains open. The condition is sufficient, not necessary: source-max can fail when evidence values have no greater downstream utility, patched heads cancel, or curvature dominates, so natural source mass alone is not a universal circuit-selection rule.

C Experimental protocol

Controlled tasks and training. Argmax retrieval requests the value paired with the largest key; flag retrieval requests the single marked value; both are order-invariant with one identifiable evidence position. Reversed decimal addition is an order-dependent contrast, and two-evidence modular sum marks two digits and requests their sum modulo ten so copying either digit is insufficient. The primary model is a four-layer pre-norm decoder transformer with width 256, eight heads, and feed-forward width 1024, trained with AdamW, batch size 512, and warmup plus cosine decay on lengths one through five. Evaluation follows a geometric ladder to $50\times$ training length, and the scale replication uses eight layers and width 512. A cell is interpreted only after training-length exact match reaches 0.8. Order-invariant cells use four seeds; the order-dependent contrast uses two. The preregistration fixes the competence gate, break-point rule, and variance-collapse check before intervention outcomes.

Measured quantities and interventions. At the answer query we record evidence mass, attention entropy, every attention norm, participation ratio, attention-output variance, and target margins. Source-max swaps the source weight with the row maximum, source-min swaps it with the row minimum, and the distractor control swaps the largest and smallest non-source weights. All conditions preserve model parameters, prompts, transformed values, the complete sorted attention row, entropy, norms, maximum, and participation ratio; the control additionally preserves source mass.

Confirmatory inference. One family of three primary contrasts is fixed in advance: controlled source-max minus distractor control, the capacity-by-assignment interaction, and natural utility-selected source-max minus matched control. Exact two-sided sign-flip tests operate on independent trained-model clusters for the controlled claims and calibration-seed clusters for natural QA, and Holm adjustment controls familywise error across the three tests. Hierarchical intervals and all other reported cells are descriptive and unadjusted unless stated otherwise.

D Scope, boundaries, and reproducibility

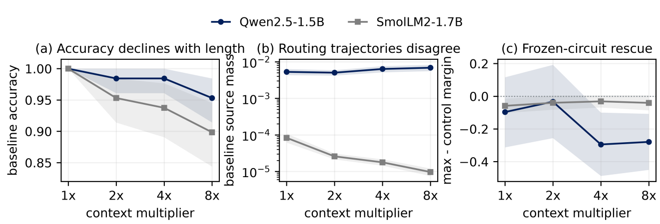

Figure 6: Matched natural-QA length ladder (seed 0; the four-passage circuit is frozen once per seed and reused at every length, and the additional Qwen seed is in the supplementary material). Bands are paired bootstrap 95% intervals over $n = 128$ examples per cell. From 4 to 32 passages, accuracy falls (a), source routing moves in opposite directions across models (b), and frozen-circuit rescue stays flat or weakens (c).

Scope. The equals-sign second-prompt format fails its competence gate in Qwen, and a blinded arrow-format pilot fails full-run competence despite directional margin signs; both are excluded from causal replications. Pythia and Qwen are floor-limited at their longest contexts, so margin is the interpretable tail outcome; Qwen-7B is quantized, and Gemma supplies a high-competence saturation boundary. The fixed-spectrum theorem still applies to the natural-QA length result because it conditions effects on transferred mass, value utility, and curvature, but the length ladder does not support the stronger claim that natural length degradation must involve source-mass loss.

Reproducibility. The controlled tasks are generated from the included task generator and require no external dataset. Natural MCQA samples are constructed from the public SQuAD training split [Rajpurkar et al., 2016] with the sampling seed and eligibility rule included. Pretrained probes use publicly released Qwen, Pythia, Gemma, and SmolLM2 checkpoints, with exact identifiers, tokenizer settings, and precision recorded in the result artifacts. Pretrained experiments used a Google Colab runtime with one NVIDIA T4 GPU, PyTorch, and Hugging Face Transformers; the earlier controlled artifacts retain configurations and seeds but not the exact package-version snapshot, so we mark infrastructure reproducibility partial. Saved artifacts reproduce every reported statistic and figure without loading a language model, and an anonymized code and artifact archive accompanies the submission as supplementary material.